

Bayesian Approach and BIC

- The Bayesian Information Criterion, also known as Schwarz criterion, is also applicable in settings where the fitting is carried out by maximization of a log-likelihood function.
- Its generic form is:

$$BIC = -2 \log \text{lik} + d \log N.$$

- For instance, consider a Gaussian model: $y_i = f(\mathbf{x}_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2) \forall i = 1, \dots, N$.
- The likelihood function writes:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^N \left(\frac{1}{\sigma \sqrt{2\pi}} \right) \exp \left(-\frac{1}{2} \frac{(y_i - f(\mathbf{x}_i))^2}{\sigma^2} \right) \\ &= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^N \exp \left(-\frac{1}{2} \frac{\sum_{i=1}^N (y_i - f(\mathbf{x}_i))^2}{\sigma^2} \right), \end{aligned}$$

and the log-likelihood:

$$\begin{aligned} l(\theta) &= -N \log(\sqrt{2\pi}) - N \log \sigma - \frac{1}{2} \frac{\sum_{i=1}^N (y_i - f(\mathbf{x}_i))^2}{\sigma^2} \\ &\Rightarrow l(\theta) \propto -\frac{\sum_{i=1}^N (y_i - f(\mathbf{x}_i))^2}{\sigma^2} \end{aligned}$$

- Let $\log \text{lik} = \max_{\theta} l(\theta)$, so that

$$\begin{aligned} -2 \log \text{lik} &= \frac{1}{\sigma^2} \sum_{i=1}^N (y_i - \hat{f}(\mathbf{x}_i))^2 = \frac{N}{\sigma^2} \cdot \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}(\mathbf{x}_i))^2 \\ &\Rightarrow -2 \log \text{lik} = \frac{N}{\sigma^2} \overline{\text{Err}} \end{aligned}$$

Yielding: $BIC = \frac{N}{\sigma^2} \overline{\text{Err}} + d \log N$

$$\Leftrightarrow BIC = \frac{N}{\sigma^2} \left[\overline{\text{Err}} + (\log N) \frac{d}{N} \sigma^2 \right]$$

Comparison with AIC

- Recall that $AIC = \overline{\text{Err}} + \frac{2}{N}d\sigma^2$.

$\hookrightarrow$ BIC is proportional to AIC, with the factor 2 replaced by $\log N$.

- Take $\log N > 2 \Rightarrow N > \exp(2) \simeq 7.4$

$\hookrightarrow$ BIC tends to penalize complex models more heavily, giving preference to simpler models in selection.

- As its name suggests, BIC is motivated by the Bayesian approach to model selection.

Bayesian model selection

- Consider a set of candidate models $\{\mathcal{M}_m\}_{m=1}^M$ with corresponding parameters θ_m . We wish to choose a “best” model from among them.
- Assume a known prior distribution for the parameters of each model:

$$\mathbb{P}(\theta_m \mid \mathcal{M}_m)$$

- Let $Z_i = \{\mathbf{x}_i, y_i\}_{i=1}^N$ denote the training data. The posterior probability of a given model is then:

$$\begin{aligned} \mathbb{P}(\mathcal{M}_m \mid Z) &= \frac{\mathbb{P}(Z \mid \mathcal{M}_m)\mathbb{P}(\mathcal{M}_m)}{\mathbb{P}(Z)} \\ &\propto \mathbb{P}(\mathcal{M}_m)\mathbb{P}(Z \mid \mathcal{M}_m) \end{aligned}$$

(Because, for a set of candidate models, $\mathbb{P}(Z)$ is a constant as it does not depend on m .)

$$= \mathbb{P}(\mathcal{M}_m) \int \mathbb{P}(Z, \theta_m \mid \mathcal{M}_m) d\theta_m$$

(averaging over all possible parameter values the model could take)

$$\Rightarrow \mathbb{P}(\mathcal{M}_m \mid Z) = \mathbb{P}(\mathcal{M}_m) \int \mathbb{P}(Z \mid \theta_m, \mathcal{M}_m) \mathbb{P}(\theta_m \mid \mathcal{M}_m) d\theta_m$$

- To compare two models, $\mathcal{M}_m$ and $\mathcal{M}_l$, we then compare the posterior odds:

$$\frac{\mathbb{P}(\mathcal{M}_m \mid Z)}{\mathbb{P}(\mathcal{M}_l \mid Z)} = \frac{\mathbb{P}(\mathcal{M}_m)}{\mathbb{P}(\mathcal{M}_l)} \cdot \frac{\mathbb{P}(Z \mid \mathcal{M}_m)}{\mathbb{P}(Z \mid \mathcal{M}_l)} \quad (*)$$

$\hookrightarrow$ If the odds are greater than 1 we choose model m , otherwise we choose model l .

- Typically, we assume that the prior over models is uniform, so that $\mathbb{P}(\mathcal{M}_m)$ is constant. What matters is the rightmost quantity of (*):

$$BF(Z) = \frac{\mathbb{P}(Z \mid \mathcal{M}_m)}{\mathbb{P}(Z \mid \mathcal{M}_l)},$$

which is called the *Bayes factor*, the contribution of the data toward the posterior odds.

- We need to approximate

$$\mathbb{P}(Z \mid \mathcal{M}_m) = \int L(\theta) \pi(\theta) d\theta$$

where $L(\theta) = \mathbb{P}(Z \mid \theta, \mathcal{M}_m)$ is the likelihood, and $\pi(\theta) = \mathbb{P}(\theta \mid \mathcal{M}_m)$ the prior.

- For large N , $L(\theta)$ becomes highly concentrated around $\hat{\theta}_m$ and we can evaluate the prior at the MLE and treat it as constant:

$$\pi(\theta) \simeq \pi(\hat{\theta}_m)$$

Laplace approximation of the marginal likelihood

- Consider a 2nd order Taylor expansion of $l(\theta)$ around $\hat{\theta}_m$:

$$\begin{aligned} l(\theta) &\simeq l(\hat{\theta}_m) + \nabla_{\theta} l(\hat{\theta}_m)' (\theta - \hat{\theta}_m) + \frac{1}{2} (\theta - \hat{\theta}_m)' \nabla_{\theta\theta'} l(\hat{\theta}_m) (\theta - \hat{\theta}_m) \\ &= l(\hat{\theta}_m) + \frac{1}{2} (\theta - \hat{\theta}_m)' \nabla_{\theta\theta'} l(\hat{\theta}_m) (\theta - \hat{\theta}_m) \end{aligned}$$

(as $\hat{\theta}_m = \arg \max_{\theta} l(\theta)$)

- Let $H = -\nabla_{\theta\theta'} l(\hat{\theta}_m)$ be the Hessian matrix, then:

$$l(\theta) \simeq l(\hat{\theta}_m) - \frac{1}{2} (\theta - \hat{\theta}_m)' H (\theta - \hat{\theta}_m)$$

- Substituting:

$$\begin{aligned} \mathbb{P}(Z \mid \mathcal{M}_m) &= \int L(\theta) \pi(\theta) d\theta \\ &\simeq \pi(\hat{\theta}) \int L(\theta) d\theta = \pi(\hat{\theta}) \int \exp(l(\theta)) d\theta \\ &\simeq \pi(\hat{\theta}) \int \exp \left[l(\hat{\theta}) - \frac{1}{2} (\theta - \hat{\theta})' H (\theta - \hat{\theta}) \right] d\theta \\ &\Rightarrow \mathbb{P}(Z \mid \mathcal{M}_m) \simeq \pi(\hat{\theta}) \exp(l(\hat{\theta})) \int \exp \left\{ -\frac{1}{2} (\theta - \hat{\theta})' H (\theta - \hat{\theta}) \right\} d\theta \end{aligned}$$

- Let $\mathbf{u} = \theta - \hat{\theta} \Rightarrow d\theta = d\mathbf{u}$ and:

$$\mathbb{P}(Z \mid \mathcal{M}_m) \simeq \pi(\hat{\theta}) \exp(l(\hat{\theta})) \int_{\mathbb{R}^{p_m}} \exp \left\{ -\frac{1}{2} \mathbf{u}' H \mathbf{u} \right\} d\mathbf{u}$$

where p_m is the number of parameters in model m .

Diagonalizing the Hessian

- Because H is the Hessian matrix, it is symmetric. It can therefore be diagonalized using an orthogonal matrix P such that:

$$H = P' \Lambda P$$

where Λ is a diagonal matrix containing the eigenvalues $\lambda_1, \dots, \lambda_{p_m}$.

- Since H is positive definite, $\lambda_j > 0 \forall j = 1, \dots, p_m$.
- Define $\mathbf{v} = P\mathbf{u}$. Because P is orthogonal, we have that:

$$\det(P) = \det(P') \quad \text{and} \quad P'P = I$$

$$\Rightarrow \det(P'P) = \det(I) = 1$$

and $\det(P)^2 = 1 \Rightarrow \det(P) = \pm 1 \Rightarrow |\det(P)| = 1$,
hence $d\mathbf{v} = d\mathbf{u}$. Now, rewrite:

$$\mathbf{u}'H\mathbf{u} = \mathbf{u}'(P'\Lambda P)\mathbf{u} = (P\mathbf{u})'\Lambda(P\mathbf{u}) = \mathbf{v}'\Lambda\mathbf{v}$$

- Because Λ is diagonal: $\mathbf{v}'\Lambda\mathbf{v} = \sum_{j=1}^{p_m} \lambda_j v_j^2$, substituting:

$$\begin{aligned} \mathbb{P}(Z \mid \mathcal{M}_m) &\simeq \pi(\hat{\theta}) \exp(l(\hat{\theta})) \int_{\mathbb{R}^{p_m}} \exp\left(-\frac{1}{2} \sum_{j=1}^{p_m} \lambda_j v_j^2\right) dv_1 \cdots dv_{p_m} \\ &= \pi(\hat{\theta}) \exp(l(\hat{\theta})) \int_{\mathbb{R}^{p_m}} \prod_{j=1}^{p_m} \exp\left(-\frac{1}{2} \lambda_j v_j^2\right) dv_1 \cdots dv_{p_m} \\ &= \pi(\hat{\theta}) \exp(l(\hat{\theta})) \prod_{j=1}^{p_m} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2} \lambda_j v_j^2\right) dv_j \end{aligned}$$

- Each factor is a standard Gaussian integral:

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2} \lambda_j v_j^2\right) dv_j = \sqrt{\frac{2\pi}{\lambda_j}}$$

so that:

$$\mathbb{P}(Z \mid \mathcal{M}_m) \simeq \pi(\hat{\theta}) \exp(l(\hat{\theta})) \prod_{j=1}^{p_m} \sqrt{\frac{2\pi}{\lambda_j}} = \pi(\hat{\theta}) \exp(l(\hat{\theta})) (2\pi)^{p_m/2} \left(\prod_{j=1}^{p_m} \lambda_j\right)^{-1/2}$$

- Because $H = P'\Lambda P$:

$$\Rightarrow \det(H) = \det(P') \det(\Lambda) \det(P) \Rightarrow \det(H) = (\pm 1) \prod_{j=1}^{p_m} \lambda_j (\pm 1) = \prod_{j=1}^{p_m} \lambda_j,$$

giving:

$$\mathbb{P}(Z \mid \mathcal{M}_m) \simeq \pi(\hat{\theta}) \exp(l(\hat{\theta})) (2\pi)^{p_m/2} \det(H)^{-1/2}$$

$$\begin{aligned} \Rightarrow \log \mathbb{P}(Z \mid \mathcal{M}_m) &= \log \pi(\hat{\theta}) + l(\hat{\theta}) + \frac{p_m}{2} \log(2\pi) - \frac{1}{2} \log(\det(H)) \\ &= \log \mathbb{P}(Z \mid \hat{\theta}_m, \mathcal{M}_m) + \log \pi(\hat{\theta}) + \frac{p_m}{2} \log(2\pi) - \frac{1}{2} \log(|H|) \end{aligned}$$

- Because $l(\theta)$ is a sum over N terms, each entry of the Hessian is a sum of N terms. Then, the entire matrix scales linearly with N , so that $H = O(N)$. Then its determinant scales as N^{p_m} , i.e. $\det(H) = O(N^{p_m})$

$$\Rightarrow \log(\det(H)) = \log(O(N^{p_m})) = p_m \log N + O(1)$$

- Substituting:

$$\log \mathbb{P}(Z \mid \mathcal{M}_m) = \log \mathbb{P}(Z \mid \hat{\theta}_m, \mathcal{M}_m) + \log \pi(\hat{\theta}) + \frac{p_m}{2} \log(2\pi) - \frac{1}{2} [p_m \log N + O(1)]$$

Recovering the BIC criterion

- Absorbing all of the constant terms:

$$\log \mathbb{P}(Z \mid \mathcal{M}_m) = \log \mathbb{P}(Z \mid \hat{\theta}_m, \mathcal{M}_m) - \frac{p_m}{2} \log N + \left[\log \pi(\hat{\theta}) + \frac{p_m}{2} \log(2\pi) - \frac{1}{2} O(1) \right]$$

$$\log \mathbb{P}(Z \mid \mathcal{M}_m) = \log \mathbb{P}(Z \mid \hat{\theta}_m, \mathcal{M}_m) - \frac{p_m}{2} \log N + O(1)$$

- If we define the loss function as $-2 \log \mathbb{P}(Z \mid \hat{\theta}_m, \mathcal{M}_m)$, we obtain an expression similar to the BIC criterion given earlier.

$\hookrightarrow$ Choosing the model with minimum BIC is equivalent to choosing the model with the largest approximate posterior probability.

- Note that BIC is asymptotically consistent as a selection criterion. This is not the case for the AIC, which tends to choose models that are too complex as $N \rightarrow \infty$.
- On the other hand, BIC often chooses models that are too simple in finite samples, because of its heavy penalty on complexity.